



India Vision 2024

DPIIT has unveiled plans to support entrepreneurs with including tax cuts.
<https://digit.in/may19-93>



Indian Startups thriving

India rose to rank 17 from 37 in StartupBlink's rankings, indicating rapid growth.
<https://digit.in/mar19-94>

tech

STARTUP CORNER

uct Flow is an all-electric scooter that boasts of some pretty impressive numbers. With a top speed of 60kmph, a range of 160km and the ability to fast charge up to 70% of its battery in one hour, all add up to a very impressive prospect. There's also regenerative braking onboard that recycles 6% of the battery capacity. However, there's a lot happening with the Flow even beyond typical automotive specifications. A number of tech features like Geo Fencing, Change in driving pattern alert, battery use analysis and even fleet management provide a user friendly experience with the Flow.

When a company files for 16 patents in the EV domain in a span of three years, it is evident that they know what they're doing. GoGreenBOV was quite successful before vehicle issues forced them to dig deep into R&D and come up with a product that was free from the drawbacks of the past. As the only Indian EV company which showcased at CES 2019, GoGreenBOV isn't shy of its innovations either. Their current focus is on B2B and hence their vehicles are not built for speed, but for carrying capacity and torque. They have major plans like putting out 10,000 vehicles by next year and 40,000 by FY21.

EVs aren't restricted to scooters. Bangalore-based EV startup EM-FLUX made a lot of waves when they demonstrated their prototype earlier this year. Capable of a 200km range,

Ather has put up some impressive specs on its bikes, and makes a strong case for EVs



RETROFITTING

Going electric doesn't always mean buying a brand new vehicle. Hyderabad-based e-Trio has recently obtained regulatory approval to retrofit petrol and diesel vehicles into electric vehicles. According to them, this transformation can be done economically in a span of 36 hours. They have already started work with retrofitting for 10 different models and plan to roll out 50 retrofitted cars for the initial customers. In fact, the first retrofitted car has already been registered by RTO Telangana. Currently, these kits can achieve 150km per charge, but the company is working on pushing that range all the way up to 230+ kms.

the superbike can crack 100kmph in just 3 seconds, and 80% of its charge in approximately 30 minutes. Other significant names in this space include Tork with the Tork T6X and Revolt Intellicorp, driven by Micromax founder Rahul Sharma.

Strom Motors has developed the Strom R3, an electric trike made in India that claims to be highly efficient. It is designed in the reverse trike format and uses roughly 30% of the parts of a typical petrol vehicle. It can crack a top speed of 80kmph and go on for 200kms. In terms of dimensions, it is smaller than a Maruti Alto and lighter as well, since all it carries are the wheels, the power motor, and

the shaft that transforms power to the wheels. Currently, Strom Motors can manufacture 5000 vehicles a year, but founder Pratik says he can ramp up production to 60,000 units per year if the EV market takes off over the next decade.

Another three wheeler startup doing great work is Gayam Motor Works (GMW) based out of Hyderabad. Founded in 2011, the startup primarily focuses on manufacturing electric auto rickshaws. They have already sold over 5000 vehicles in 15 countries and are also working on high-efficiency battery swapping technology. According to GMV, their vehicles operate on 1/7th the cost of petrol vehicles. Multiple major clients like Flipkart, BigBasket, Grofers and even IKEA have incorporated GMV vehicles into their fleets.

ENERGY INFRASTRUCTURE

Any number of options in the electrical vehicle sector is not good enough if the requisite infrastructure to support it doesn't exist. While a number of startups mentioned earlier are trying to set up their own networks of charging stations, some are focussed on solving the issues in mobility that other startups can rely on. In fact, one of the biggest issues causing sluggish EV adoption in the country is the lack of standardisation in the charging ecosystem. These startups might change that if they can achieve their targets.

Mumbai-based ION Energy is working on developing a platform that optimises EV battery usage, using software analytics. The companies working with them would still be making their own batteries, licensing ION Energy technology. Along with the battery management system, which helps companies monitor and build better batteries, ION has also unveiled intelligent, modular, and portable 48V lithium-ion battery. SUN Mobility is another startup working in the area that provides a replaceable battery solutions where they maintain ownership of the battery, freeing vehicle owners from battery issues. 